Monitoring report — run T20000

Legend: **[measured]** = figure read from data; **[heuristic]** / **[hypothesis]** = unverified interpretation; **[conjecture]** = unproven result.

Turing–Backlund validation: □ COMPLET [measured, read from execution_v16_T20000_20261025_170038.log] — numerical check, not a certification.

1. Summary

Item	Value
Run	20261025_170037_T20000 (prod)
Start / end	2026-10-25 17:00:37 → 17:01:20
Duration	0min43s
Command (excerpt)	`bash -c printf '20000\n0\n' | python /home/riemann/projet_zeta/src/calculs/optimisation/co`
Exit code	0
CPU turbo start → end	actif (intel_pstate) → actif (intel_pstate)
Zeros CSV	`zeros_v16_T20000_20261025_170038.csv`
Turing–Backlund	□ COMPLET
Zeros in CSV	22491 [measured]
t min → t max	14.135 → 19999.276
Average rate (zeros / run duration)	523.0 /s

2. Machines during the run

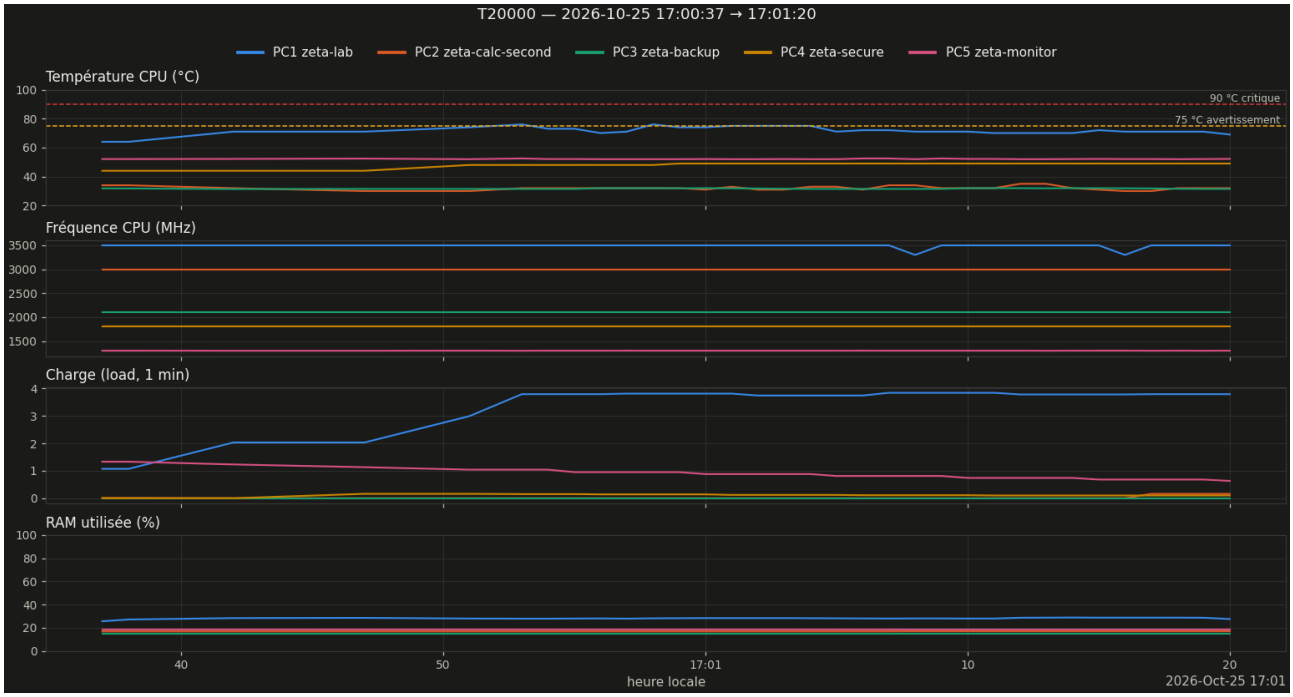


Figure 1 — Temperature, frequency, load and RAM of the 5 machines (1 sample/s).

Machine	Valid	T moy °C	T max °C	% ≥ 75 °C	MHz min–max	load max	RAM max %
PC1 zeta-lab	100 %	71.6	76.0	18	3300–3500	3.84	28.8
PC2 zeta-calc-second	100 %	32.1	35.0	0	2992–2992	0.16	16.9
PC3 zeta-backup	100 %	31.7	32.0	0	2100–2100	0.00	15.2
PC4 zeta-secure	100 %	48.2	49.0	0	1800–1800	0.16	—
PC5 zeta-monitor	100 %	52.1	52.5	0	1297–1299	1.33	18.6

Observations

- PC1: max temperature 76.0 °C, 18 % of time ≥ 75 °C (warning threshold) [measured]

3. Computed zeros

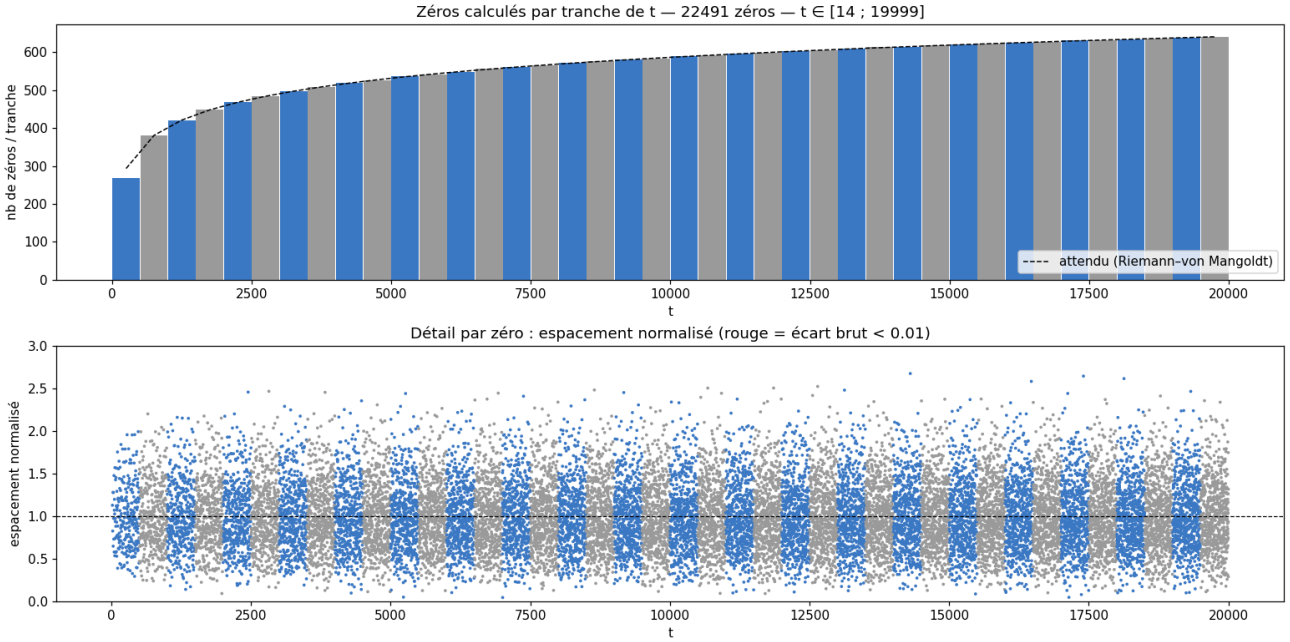


Figure 2 — Top: zeros per t-slice (alternating blue/grey) vs theoretical density ln(t/2π)/2π. Bottom: normalized spacing of each zero.

Count: 22491 zeros in the CSV; Riemann–von Mangoldt formula (without the S(t) term): ≈ 22491.0. Difference -0.0. **[heuristic]** a difference of a few units is normal; tens suggest missing or duplicated zeros. **This is not a validation: completeness is checked by Turing–Backlund (verdict at the top of this report).**

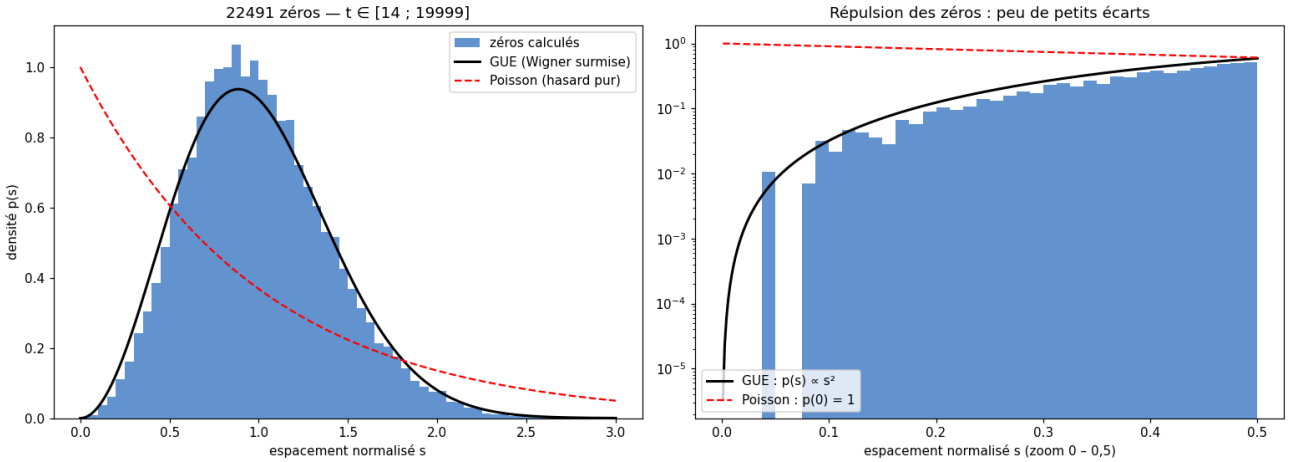


Figure 3 — Normalized spacings vs GUE (Wigner surmise) and Poisson; right: zoom on repulsion (log scale).

Item	Measured	Reference
Mean of s	1.0000	1 (by construction)
Variance of s	0.1569	GUE ≈ 0.18 ; Poisson = 1
Share s < 0.1	0.062 %	GUE surmise ≈ 0.11 % ; Poisson ≈ 9.5 %
Smallest raw gap	0.035307	—

[conjecture, Montgomery–Odlyzko] spacings follow GUE; here the repulsion is clear. A variance slightly below 0.18 at moderate heights is attributed to finite-size corrections **[hypothesis]**. With few zeros (< a few thousand) the histogram is noisy: do not over-interpret.

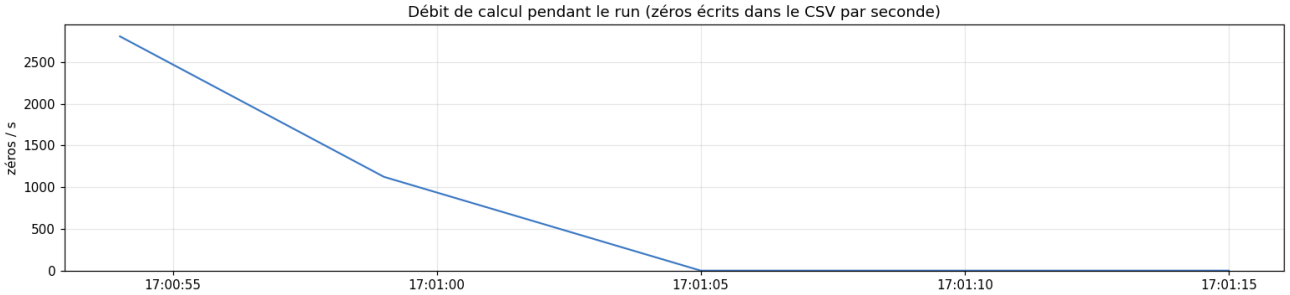


Figure 4 — Throughput (zeros/s) over the run, to compare with PC1 frequency and temperature (figure 1).

- Instantaneous throughput not usable: the zeros CSV grew on only 40 % of the samples (written per worker at the end); rely on the mean throughput in the summary [measured]

4. Session sheet

Item	Value
CPU	Intel(R) Core(TM) i7-7500U CPU @ 2.70GHz (4 threads)
Governor / turbo	performance / actif (intel_pstate)
Git	72bb991 @ Riemann_Lab_IA (35 modified files)
Load / free RAM at start	1.07 / 11.6 Go
Detailed file	`session.md`

5. Interpretation

Written by Claude Code from the figures above — to be validated.

- **[mesuré]** 22 491 zéros, validation Turing–Backlund COMPLET, écart à la formule de Riemann–von Mangoldt : −0,0.

- **[mesuré]** Durée suivie 43 s ; `time` 52 s : l'écart de 9 s est identique sur quatre runs (T=20000 et T=200000).
- **[mesuré]** PC1 à 75 °C ou plus pendant 18 % du run (maximum 76 °C) ; PC2 à PC5 inactives.
- **[mesuré]** Variance des espacements normalisés 0,1569 (GUE : environ 0,18) ; 0,1614 pour T=100 000 et 0,1693 vers t = 5·10^6. **[hypothèse]** effet de hauteur finie, non démontré.
- **[hypothèse]** Ce run est 16 % plus long que celui du 24/10 22:56 (43 s contre 37 s) alors que sa température maximale est plus basse (76 °C contre 77 °C) : la température seule ne l'explique pas ; cause inconnue.

6. Limits

- Samples about every 1 s (SSH): shorter spikes are missed; PC4 (OpenBSD) has no RAM reading.
- Run start/end = detected process appearing/disappearing, within a few seconds.
- Beyond 8,000 points per machine the plot is subsampled (statistics use all samples).